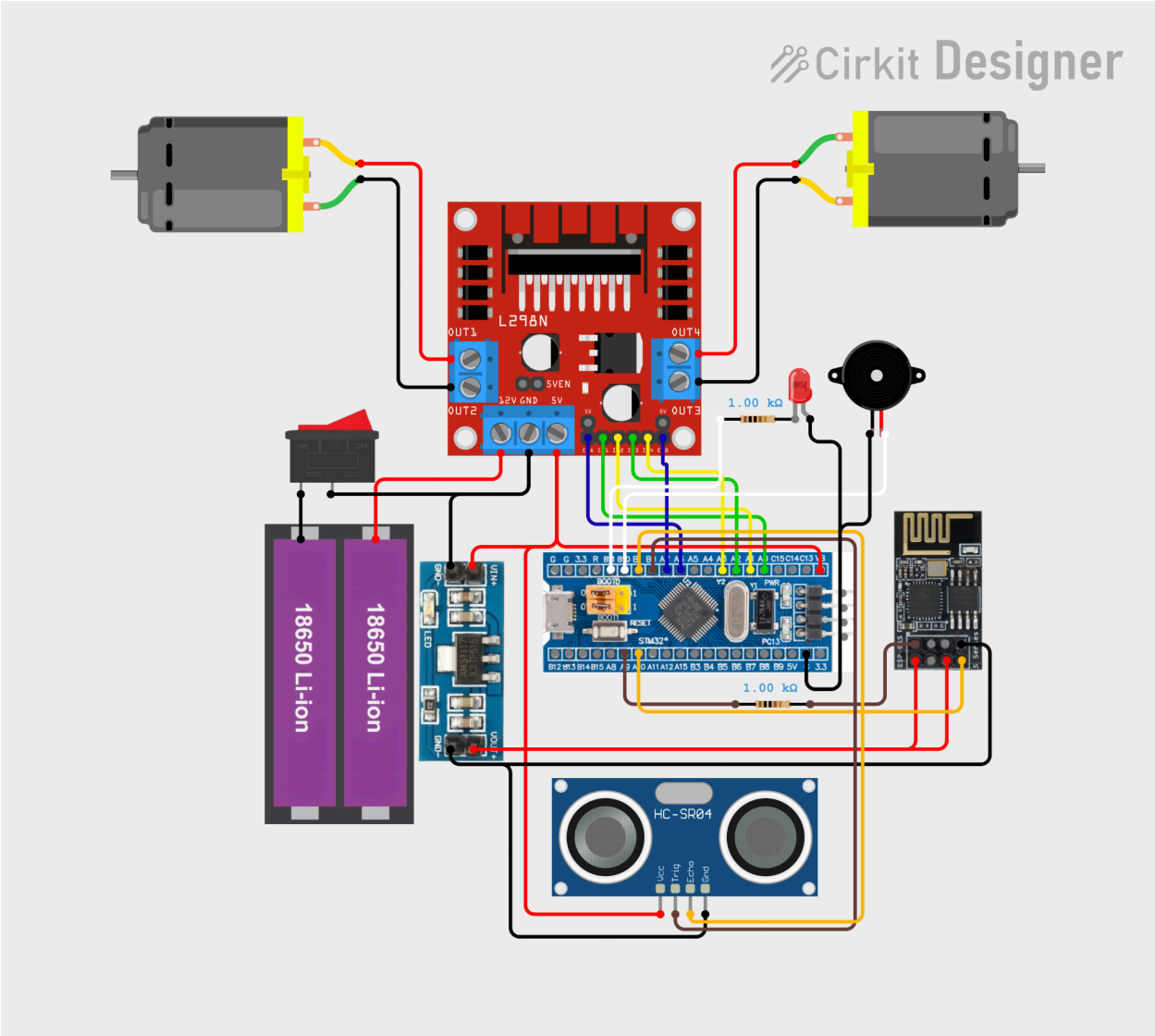
Guard-X:

Embedded V2V Collision Detection and Prevention System

Guard-X is an innovative Vehicle-to-Vehicle (V2V) early collision detection and prevention system developed using STM32 microcontrollers. The system continuously gathers real-time data on distance and speed through sensors like ultrasonic or radar modules. It communicates this information wirelessly between nearby vehicles to predict possible collisions and alert drivers through visual or audio warnings. By enabling intelligent inter-vehicle communication and quick driver response, Guard-X aims to significantly enhance road safety, reduce accident risks, and move toward smarter transportation systems.

Circuit Diagram



Circuit Documentation

Summary

This circuit is a microcontroller-based system designed to control two DC motors using an L298N motor driver. It also includes a WiFi module for connectivity, an ultrasonic sensor for distance measurement, a buzzer for audio alerts, and an LED indicator. The system is powered by two 18650 Li-ion batteries, with a 3.3V converter to supply power to components requiring lower voltage.

Component List

1. **STM32F103C8T6 Microcontroller**
    - A versatile microcontroller used for controlling the entire system, interfacing with sensors, and managing motor operations.
2. **DC Motors (2 units)**
    - Actuators used for mechanical movement, controlled via the L298N motor driver.
3. **L298N DC Motor Driver**
    - A dual H-bridge motor driver that controls the direction and speed of the DC motors.
4. **WiFi Module ESP8266-01**
    - Provides wireless connectivity for remote control and data transmission.
5. **Rocker Switch**
    - A simple on/off switch used to control the power supply to the circuit.
6. **18650 Li-ion Battery x 2**
    - Provides the main power supply for the circuit.
7. **Resistors (2 units, 1000 Ohms each)**
    - Used for current limiting and voltage division in the circuit.
8. **3.3V Converter**
    - Converts the battery voltage to 3.3V for components that require lower voltage.
9. **HC-SR04 Ultrasonic Sensor**
    - Measures distance by emitting ultrasonic waves and calculating the time taken for the echo to return.
10. **LED: Two Pin (green) - Long Pins**
    - Provides visual indication of the system status.
11. **Buzzer**
    - Emits sound for audio alerts or notifications.

Wiring Details

STM32F103C8T6 Microcontroller

- **A0**: Connected to L298N IN1
- **A1**: Connected to L298N IN2
- **A2**: Connected to L298N IN3

- **A3**: Connected to L298N IN4
- **A6**: Connected to L298N ENA
- **A7**: Connected to L298N ENB
- **A9**: Connected to Resistor (1000 Ohms) pin1
- **A10**: Connected to ESP8266-01 TX
- **B0**: Connected to HC-SR04 TRIG
- **B1**: Connected to HC-SR04 ECHO
- **B10**: Connected to Buzzer VCC
- **B11**: Connected to Resistor (1000 Ohms) pin1
- **G**: Connected to LED anode and Buzzer GND
- **VBAT**: Connected to L298N 5V

DC Motors

- **Motor 1**:
  - **pin 1**: Connected to L298N OUT2
  - **pin 2**: Connected to L298N OUT1
- **Motor 2**:
  - **pin 1**: Connected to L298N OUT4
  - **pin 2**: Connected to L298N OUT3

L298N DC Motor Driver

- **OUT1**: Connected to Motor 1 pin 2
- **OUT2**: Connected to Motor 1 pin 1
- **OUT3**: Connected to Motor 2 pin 2
- **OUT4**: Connected to Motor 2 pin 1
- **12V**: Connected to Battery +
- **GND**: Connected to Rocker Switch pin 2 and 3.3V converter GND
- **IN1**: Connected to STM32F103C8T6 A0
- **IN2**: Connected to STM32F103C8T6 A1
- **IN3**: Connected to STM32F103C8T6 A2
- **IN4**: Connected to STM32F103C8T6 A3
- **ENA**: Connected to STM32F103C8T6 A6
- **ENB**: Connected to STM32F103C8T6 A7
- **5V**: Connected to HC-SR04 VCC, 3.3V converter Vin, and STM32F103C8T6 VBAT

WiFi Module ESP8266-01

- **+3V3**: Connected to 3.3V converter Vout
- **CH-PD Chip power down**: Connected to 3.3V converter Vout
- **GND**: Connected to HC-SR04 GND and 3.3V converter GND
- **RX**: Connected to Resistor (1000 Ohms) pin2
- **TX**: Connected to STM32F103C8T6 A10

Switch

- **1**: Connected to Battery -
- **2**: Connected to L298N GND

18650 Li-ion Battery x 2

- **+**: Connected to L298N 12V
- **-**: Connected to Rocker Switch 1

Resistors

- **Resistor 1 (1000 Ohms)**:
  - **pin1**: Connected to STM32F103C8T6 A9
  - **pin2**: Connected to ESP8266-01 RX
- **Resistor 2 (1000 Ohms)**:
  - **pin1**: Connected to STM32F103C8T6 B11

3.3V Converter

- **GND**: Connected to L298N GND, Rocker Switch 2, and ESP8266-01 GND
- **Vout**: Connected to ESP8266-01 +3V3 and CH-PD Chip power down
- **Vin**: Connected to L298N 5V

HC-SR04 Ultrasonic Sensor

- **VCC**: Connected to L298N 5V
- **TRIG**: Connected to STM32F103C8T6 B0
- **ECHO**: Connected to STM32F103C8T6 B1
- **GND**: Connected to ESP8266-01 GND and 3.3V converter GND

LED:

- **anode**: Connected to STM32F103C8T6 G
- **cathode**: Not connected in the provided net list

Buzzer

- **VCC**: Connected to STM32F103C8T6 B10
- **GND**: Connected to STM32F103C8T6 G